

REM Sleep Disturbance and Role of Noradrenaline in Post-traumatic Stress Disorder:

A Narrative Review

Abstract

PTSD, characterised by a complex psychopathology, is a long-term effect of traumatic experiences that evoke great fear and helplessness. Sleep disturbances constitute an important feature of PTSD, viewed primarily in the form of insomnia and nightmares. In fact, the vitality of sleep in maintaining our sense of wellbeing has made it conceivable that sleep deprivation is itself a stressor to the human body and should relate to stress system activation. Research has also described sleep disturbance as a predisposing, precipitating and perpetuating component in the clinical course of PTSD. Sleep is classified as either rapid eye movement sleep (REMS) or non-REMS. For practical reasons, most experimental research has been undertaken by causing REMS loss. It has been demonstrated that withdrawal of noradrenaline (NA) is required for the formation of REMS. Furthermore, when REMS is lost, NA levels rise in the brain, and this higher NA is responsible for many of the sleep loss-related symptoms. The purpose of this review is 1) to recapitulate the available evidence that links sleep disturbances in PTSD in a complex multidirectional relationship, 2) to emphasize the critical role of NA in mediating REMS-loss associated PTSD outcomes, and 3) to summarize the NA-targeting therapeutic strategies adopted to treat REMS disturbances in PTSD.

Keywords: Sleep disturbances, PTSD, REMS loss, Noradrenaline

Introduction

The concept of psychological trauma has become substantially crucial to the scientific community on a global scale. Commonly defined as a response to an extremely stressful event(s), trauma can be a consequence of exposure to violence, disasters, terrorism, war, illness, and other overwhelmingly disturbing events. Its prevalence across borders is concerning since trauma raises the likelihood of developing a complex psychopathology, stemming from a cascade of neuroendocrinological, neurochemical, and neuroanatomical changes (Bowirrat et al., 2010). The neurobiological impression of trauma involves acute stress responses initiated and maintained by the neural circuitry (Gradus & Galea, 2023), as noted below.

The information of a traumatic event acquired by the amygdala, the initial responder, is quickly evaluated for danger, with the potential to activate the pituitary gland, which ultimately releases stress hormones. The hippocampal memory system aids in this evaluation by providing information from its database of previous threats to the amygdala. This appraisal, in conjunction with the prefrontal cortex (important for emotional control and decision-making), is key in deactivating the hypothalamus-pituitary-adrenal (HPA) axis when the threat has passed. Upon being alerted by the amygdala, the HPA axis releases chemicals (neurotransmitters) and hormones that promote survival by mobilizing or enhancing life-sustaining systems, all of which return to pre-threat status once the real/perceived threat has subsided (Kumar et al., 2013; Lupien & McEwen, 1997; Plotsky, 1987). Depending on the severity, frequency, and duration of traumatic event, the HPA axis may continue to flood the body with cortisol consequently stimulating the hyperactivity of the sympathetic nervous system, combined to create a toxic stress effect in the bodily systems (Bremner & Wittbrodt, 2020; McEwen, 2017). Hyperarousal, an abnormally heightened physiological response to a stressful stimulus, is a consequence of the neurobiological shifts triggered by trauma. Although it acts to promote a higher possibility of survival by inducing a constant state of vigilance, other capacities may be compromised.

Hyperarousal can impair an individual's ability to perceive and respond effectively to specific stimuli, such as loud noises or abrupt movements, and sometimes cause over-reactions to situations that appear threatening when they are not. A trait characteristic of hyperarousal, sleep reactivity, manifests as sleep disruption (SD) (Kalmbach et al., 2018). Therefore, hyperarousal, or, more specifically, sleep disruption, is a crucial area for neuroscientific exploration.

Sleep can elevate emotions and lessen the emotional impact of painful memories. People with insomnia disorder, or psychological disorders characterized by significant sleep disruptions, will be the first to admit that their attempts to initiate sleep are typically followed by interruptions and frustrations, which rarely ease the emotional distress they experienced during the previous day (K. E. Miller & Gehrman, 2019). Sleep is broadly divided into rapid eye movement sleep (REMS) and non-REM sleep (NREMS). REMS is where one spends the least time and encompasses rapid movements of the eye, increased brain activity, and loss of muscle tone (Roffwarg et al., 1966). Dreaming also occurs during this phase. NREMS, or the slow wave, deep sleep precedes the appearance of REMS and is a prerequisite for REMS to occur. NA is a crucial biomolecule which lowers during REMS, and its rise is noted in sleep deprivation (R. Mehta, Singh, Khanday, et al., 2017). The release of NA from the locus coeruleus (LC) suppresses the activity of brain areas implicated in REMS promotion, such as pontine cholinergic neurons. NA has implications for most REMS-linked abnormalities, making it the most targeted neurotransmitter underlying explanation and design of therapeutics to treat SD (R. Mehta et al., 2020).

Although it is now widely acknowledged that sleep, in general, is essential for memory consolidation, REMS and NREMS appear to have distinct purposes. The disruption of REMS appears to preferentially consolidate emotional memories through a reverse learning process to help an individual cope with aversive situations, thus being associated with nightmares as well as control of muscle tone activity (Suchecki et al., 2012). Apart from NA, other

neurotransmitters operate in an orchestrated manner to regulate REMS (Fraigne et al., 2015). Serotonin reduces REMS formation and determines the dream's substance, vividness, and emotional tone (Iwasaki et al., 2018), while dopamine has implications in regulation of REMS (Zhang et al., 2022). GABAergic neurons in the brainstem inhibit motor activity and suppress muscle activity during REMS (Luppi et al., 2006), while REM-On neurons release glutamate, which stimulates brain regions responsible for motor output, emotional processing, and memory (Mondino et al., 2021a).

SD is very common in trauma-exposed individuals that can come in the form of early awakening, restless sleep, difficulty falling asleep, and nightmares, stemming from neurotransmitter-related dysregulation of arousal systems (Murkar & De Koninck, 2018). Numerous strategies may have been deployed for the immediate intervention of acute stress symptoms including medication, good sleep hygiene practices, cognitive rehearsals of, nightmares, relaxation strategies, and nutrition (Wagner et al., 2006). However, SDs sometimes remain resistant to intervention long after other traumatic stress symptoms are successfully treated. This persistence in the occurrence of sleep-disruptive mechanisms in turn interferes with the resolution of harmful stress responses and prevents a considerable minority from bouncing back with resiliency (Garthus-Niegel et al., 2015). Inability to emotionally regulate could pave way for one of the most prevalent anxiety disorders, Post Traumatic Stress Disorder (PTSD) (Colvonen et al., 2019; Vanderheyden et al., 2015), also referred to by Koenan et al (2017, p. 2) as a "life sentence due to its link with higher risk of chronic disease, hastened ageing, and premature mortality."

PTSD is most famously recognized and hence defined as “recurrent, unwanted mental and emotional re-experiencing of a previous traumatic event”, wherein trauma can be life-threatening or life-altering. In PTSD, overall symptoms are usually characterized by re-experiencing, i.e., nightmares and daytime flashbacks; negative cognitions and mood;

avoidance and numbing, i.e., escape from reminders of the traumatic event; hyperarousal, as viewed in the form of insomnia (Miao et al., 2018; Sareen, 2014). The prevalence of this disorder ranges from approximately 12 months (2-3% cases) to a lifetime of persistent symptoms (7-8% cases) and is reportedly observed twice as many times in women than in men (Christiansen & Hansen, 2015; van der Kolk, 2000). Traumatic memories in PTSD are caused, in part, by a failure in extinction learning, which is the learning that the previously conditioned stimuli no longer represent a danger. It is widely hypothesized that REMS abnormalities caused by NA hyperactivity associated with PTSD impede the consolidation of extinction memory, resulting in the inability of the extinction memory to endure and generalize (Kung et al., 2012; R. Mehta et al., 2020; Mellman et al., 1995a; Vanderheyden et al., 2014). So far, experimental support for this approach is finite, but rising. Other prominent characteristics such as decreased slow wave sleep, elevated REMS density, and REMS fragmentation, are also suggested to result from patients with elevated NA-ergic tone during REMS (Germain et al., 2008; R. Mehta et al., 2020). As part of both the sleep-wake and PTSD-related circuits, hyperactive NA-ergic projections from the LC might represent a final shared pathway in creating a state of hyperarousal (Kim et al., 2013; Kung et al., 2012; Mellman et al., 1995a; Raskind et al., 2013).

In this review, we will first elaborate on the multifaceted relationship reported between SD and PTSD. We also attempt to recapitulate available evidence on various neurochemical dysregulations regularly observed in PTSD and how they may be linked to the REMS disruption, complicating its further clinical course. Although, the variability in REMS in case of PTSD involves a complex interplay of multiple neurotransmitters and brain regions noted here, we will emphasize, firstly and more importantly, on the role of NA in associating these two phenomena, given its prominence in use throughout the decades to design suitable interventions. Further, we also study how factors like ageing, gender and gene-environment interactions may

influence these variations, along with comparing the crosstalk in some other associated disorders.

Sleep disruption in PTSD - A Vicious Cycle

Sleep difficulties are common in PTSD, and 70-90% of patients experience them. Most PTSD patients (around 50-70%) report recurrent nightmares, which primarily occur during REMS. Periods of sleep paralysis, which often occur during (REM) sleep-wake transitions, are also relatively prevalent in PTSD. SD is not merely a symptom or a result of PTSD acquisition; rather, it is strongly suggested as a predisposing, precipitating and perpetuating factor for PTSD. Sleep difficulties immediately following trauma due to aforementioned acute stress response to traumatic events raise the likelihood of PTSD acquisition (Brownlow et al., 2016). Using studies conducted on patients with both pre-trauma sleeplessness and dreams, results have strongly pointed out that pre-traumatically existent insomnia also governs the occurrence of PTSD in such cases (Bryant et al., 2010) (Koffel et al., 2013) (H. E. Wang et al., 2019).

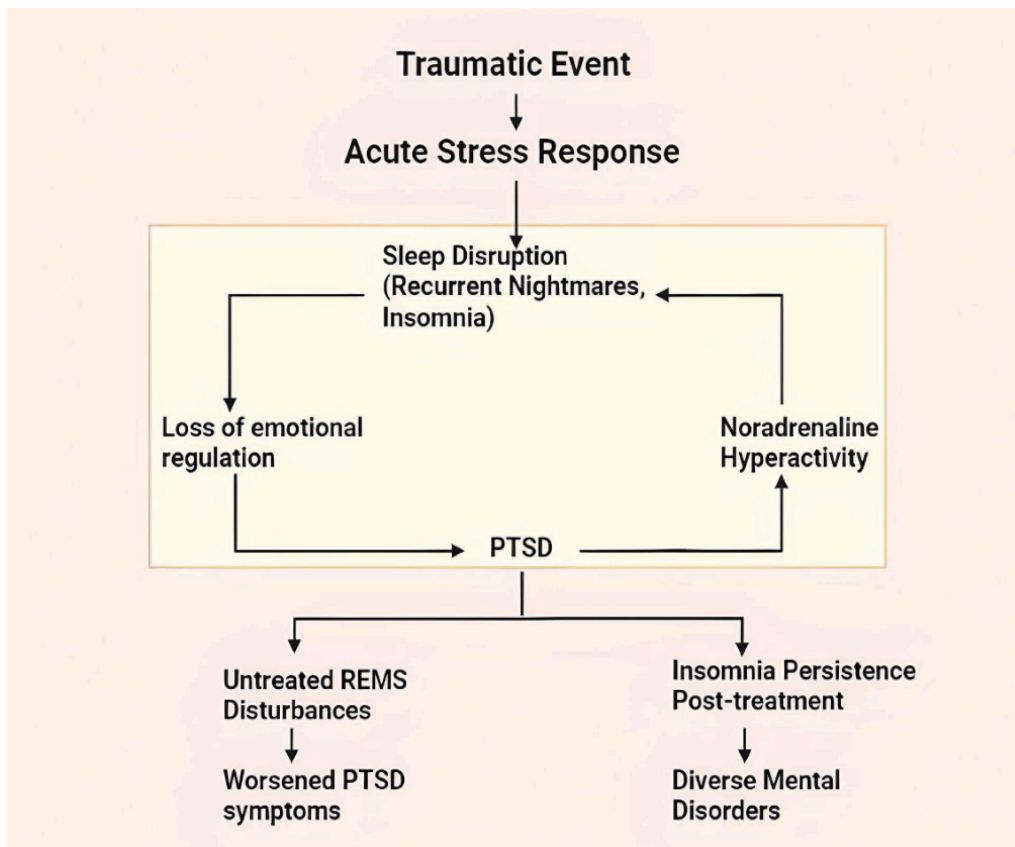


Figure 1.

Exposure to a TE induces a series of stress responses, including sleep disturbances like insomnia and recurrent nightmares (Murkar & De Koninck, 2018). Unresolved sleep problems/chronic insomnia results in an inability to regulate emotional processing, thus predisposing a person to developing PTSD (Brownlow et al., 2016). The neurobiological changes instigated as a response to PTSD prompts NA hyperactivity (Blanchard et al., 1991a), which further worsens the REMS quality, altering the clinical course of the disorder (R. Mehta, Singh, & Mallick, 2017).

Acting as a causal factor, sleep disruptive mechanisms of PTSD also alter its clinical course, as they may stay as comorbid disorders with a life of their own. For instance, a study of 108 active-duty army soldiers, a group frequently exposed to traumatic workplace events (Pruiksma et al., 2016), found that even after successful treatment for PTSD, insomnia complaints remained in 77% of the sample (Pruiksma et al., 2016). Other studies with similar results have concluded the ‘residual insomnia’ as an important prospective factor which may later lead to development and/or relapse into diverse mental disorders (Larsen et al., 2019) (Belleville et al., 2011).

On the other hand, a clinically substantial reduction in PTSD symptoms during the day may result from the successful treatment of sleep disturbances and nightmares. As a result, therapies that help patients sleep better are probably of great therapeutic relevance because they may immediately alleviate the symptoms of PTSD (Galovski et al., 2016). Thus, there exists a multidirectional relationship between sleep disruption and PTSD, wherein, both the phenomena may trigger and exacerbate each other, resulting in a vicious cycle.

REMS Loss associated PTSD: Role of NA

The brain regions associated with sleep/wake regulation and anxiety exhibit significant interactions, particularly the close relationship between the cerebral cortex and the arousal centre. This interconnectedness not only provides an anatomical basis for the cognitive hyperarousal that may link anxiety to SD but also establishes a concrete physiological pathway for their interconnectedness. The principal NA producer in the central nervous system, the LC, plays a critical role in controlling sleep-wake cycles and anxiety-related activities through its connections with cerebral cortex and amygdala (Richards et al., 2020).

Strong evidence from previous studies supports the notion of sympathetic system (NA-ergic) hyperreactivity in individuals with PTSD, which correlates with symptoms of hyperarousal and reliving, such as trauma-related dreams, flashbacks, intrusive memories, and

emotional and physical responses to traumatic stimuli (Blanchard et al., 1991a; Liberzon et al., 1999a; Mellman et al., 1995b; Pitman et al., 2012a; Southwick et al., 1993a). Stress-induced sympathetic nervous system activation triggers secretion of NA and adrenaline from the adrenal medulla. This surge in the NA release from sympathetic nerve endings, along with changes in blood flow to various organs as part of the stress response, is mediated through postsynaptic $\alpha 1$, $\beta 1$, and $\beta 2$ receptors, while the presynaptic autoreceptor $\alpha 2$ inhibits NA release (Southwick et al., 1999). Studies have consistently reported significantly higher NA levels in individuals with PTSD than controls, supporting the theory that NA plays a role in the pathophysiology of PTSD (Mellman et al., 1995b). The increase in NA-ergic signaling may stem from elevated NA release, altered brain responses to NA, or a combination of both factors (Hendrickson et al., 2018).

Reduced levels or affinity of alpha-2-adrenergic autoreceptors, along with the neuropeptide Y (NPY), which decreases NA release, could contribute to the increased release of NA in response to sympathetic system activation in PTSD (Perry et al., 1987; Rasmusson et al., 2000). The secretion of the neuropeptides on LC alters the NA release which may impact the single cells to network throughout the brain (Illes et al., 1993; H. S. Lee et al., 2005). Lower plasma NPY levels have been linked to slower healing of PTSD symptoms over time, according to a retrospective study of male veterans (Yehuda et al., 2006).

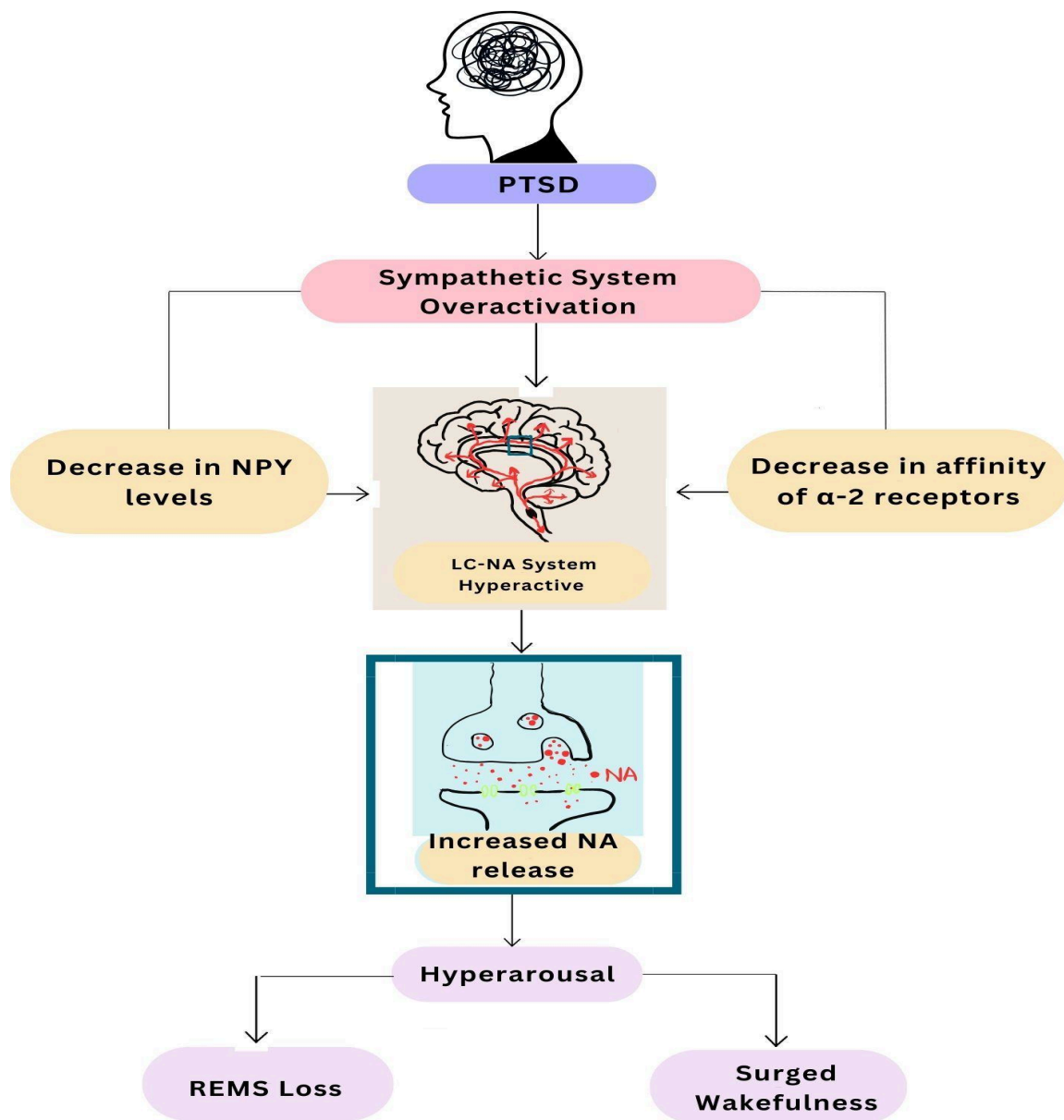


Figure 2.

Hyperactivation of sympathetic system in response to chronic stress in (Bremner & Wittbrodt, 2020; McEwen, 2017) Post traumatic stress disorder (PTSD) associates with (i) lower level of Neuropeptide Y (NPY) and (ii) decreased affinity to alpha-2 autoreceptors resulting in heightened Locus Coeruleus-Noradrenaline (LC-NA) system activity (Perry et al., 1987;

Rasmusson et al., 2000). A net resultant surge Noradrenaline (NA) level in the brain causes hyperarousal state contributing to increased wakefulness and reduces REMS.

Notably, NA-ergic neurons exhibit dormancy during REMS, whereas NA levels in the brain increases during REMS loss, thereby contributing to the acute and long-term consequences associated with REMS deprivation.(R. Mehta et al., 2016) The LC-NA system is a leading source for controlling arousal during sleep for the following reasons: Inversely associated with sleep spindle activity, LC activity increases soon before sleep-to-wake transitions, and stimulating the LC causes quick waking from sleep. (Aston-Jones & Bloom, 1981; Carter et al., 2010; Kjaerby et al., 2020; Matosevich & Nir, 2021; Southwick et al., 1993a) Optogenetic stimulation of LC neurons reduced REMS and improved wakefulness, according to studies. As a result, the same neuromodulator pathways also play a role in the link between PTSD and SD, with (hyperactive) LC-NAergic projections acting as the last shared pathway in the development of the hyperarousal state that characterizes both PTSD and disturbed sleep.

Overall, this crosstalk between NA signalling, SDs, and anxiety states highlights the interconnected nature of PTSD and disrupted sleep. The hyperactive LC-NA-ergic projections act as a shared pathway in the development of the hyperarousal state observed in both PTSD and disturbed sleep. Understanding these neurobiological mechanisms can provide insights into the development of targeted therapeutic interventions for individuals with PTSD and SDs.

Other neurobiological correlates of REMS loss and PTSD

Under the influence of PTSD, neurobiological shifts such as changes in limbic system functioning, HPA axis activity with variable cortisol levels, and neurotransmitter-related dysregulation of arousal and endogenous opioid systems occur, increasing the probability of engaging in risky behaviours and the onset of physical health issues. Recap of literature uptill here has established the crucial part of NA-control over REMS changes observed in the cases of PTSD owing to its recorded surged levels. The conjunction of reduced NPY levels as well as lowered affinity to alpha-2 autoreceptors contribute to heightened LC-NA system activity resulting in an escalated NA levels. We will further discuss the underlying

neuroendocrinological and neuroanatomical system dysfunction, followed by an elaborate review of various other neurotransmitter imbalances caused by PTSD, and their possible link to REMS disruption.

The Influence of HPA axis and brain circuits

The neuroendocrinological pathways contributing to PTSD are highly intricate and challenging to decipher. The HPA axis plays a central role in regulating stress response. It begins in the hypothalamus with the release of corticotropin-releasing hormone (CRH), which stimulates adrenocorticotrophic hormone (ACTH) secretion from the pituitary gland resulting in the release of glucocorticoids, such as cortisol (Sherin & Nemeroff, 2011a; Smith & Vale, 2006). Cortisol exerts a negative feedback control over the HPA axis, suppressing the NA-ergic stress response. Elevated levels of CRH have been observed in the cerebrospinal fluid (CSF) and blood of various cohorts of individuals with PTSD. These findings suggest that disruption of the HPA axis may contribute to the psychopathology of PTSD. Furthermore, cortisol, the product of HPA axis activation, has both anxiogenic physiological and behavioural effects. Interestingly, cortisol levels in individuals with PTSD exhibit considerable variability. Some studies have reported initially elevated cortisol levels, indicating an overly active stress response (Leproult et al., 1997). However, other studies have found low cortisol levels, particularly in individuals who have experienced recurrent interpersonal trauma or abuse (Vgontzas et al., 1999). It is important to note that factors such as sleep, exercise, and medication use can influence cortisol levels throughout the day, making it challenging to rely solely on cortisol levels for predicting the development of PTSD (Brun et al., 1998).

In conclusion, the neuroendocrine pathways involved in PTSD, particularly the HPA axis, contribute to the complex pathophysiology of this disorder. Further research is needed to fully understand the complexity of these pathways and their roles in the development and maintenance of PTSD symptoms. Furthermore, investigations in male rats have found that

pharmacological restriction of corticosterone production during REM sleep deprivation impaired sleep rebound, implying that activation of the HPA axis during REM-sleep suppression is required for normal sleep recovery. The link between REM sleep and stress-related behaviour has been emphasised further by research indicating that, in addition to elevated corticosterone levels, specifically REM sleep deprivation causes increased anxiety and depressive-like conduct in rats (Nollet et al., 2020). All these studies highlight the critical consequences of sleep deprivation, particularly REMS suppression, on physiological and behavioural stress coping systems (Chrousos & Gold, 1992; Gillin et al., 1972; Lo Martire et al., 2020).

Extensive brain imaging techniques have displayed structural alterations in neuroanatomical circuits associated with adaptations to stress and fear conditioning in brain areas including hippocampus, amygdala, and ventromedial prefrontal cortex (vmPFC). Extreme stress post-trauma causes an increase in the release of glucocorticoids, excitatory amino acids, serotonin, and other neurotransmitters and neuropeptides that may be linked to hippocampus damage (Bremner et al., 1995). Increased cortisol secretion is also known to impair hippocampal neurogenesis and damage hippocampal neurons, which might explain the smaller hippocampal size found in patients with PTSD.

It has been theorized that the vmPFC inhibits the amygdala and alteration, in this inhibition could account for the symptoms of PTSD. Of note, this has been supported by the functional imaging studies, which exhibit hypoactivity in the vmPFC but hyperactivity in the amygdala, resulting in dysregulation of negative emotions (Koenigs & Grafman, 2009). These mechanisms pose both risk factors for persistence, and consequences of PTSD. Furthermore, various studies correlating the potential association between neuroanatomical morphology and REMS symptoms in insomnia, could pose a probable underlying mechanism governing the development of REMS disorders in PTSD, or rather, development of PTSD from pre-existing disruptive REMS patterns.

Other neurotransmitter imbalances in PTSD

Noradrenaline (NA) acts as a central and peripheral modulator of autonomic stress response (Godoy et al., 2018). The sympathetic nervous system is activated by the central nervous system, which causes the release of NA and adrenaline as well as changes in blood flow to organs that support fight-or-flight behaviour. Due to its several functions in controlling arousal and autonomic stress responses, NA has been a major focus of studies into the pathophysiology of PTSD (Sherin & Nemeroff, 2011b). Exaggerated adrenergic response to the traumatic experience may influence the establishment of the persistent traumatic memories that characterise the disease (Pitman, 1989; Pitman et al., 2012b). NA-ergic transmission plays a vital role in arousal regulation and, together with glucocorticoids, in memory processing (Sherin & Nemeroff, 2011b). In PTSD patients, NA-ergic hyperreactivity is linked to hyperarousal and reliving symptoms, including as nightmares, flashbacks, intrusive memories, and emotional and physical responses to traumatic triggers (Blanchard et al., 1991b; Liberzon et al., 1999b; Mellman et al., 1995c; Southwick et al., 1993b). LC neurons fire in a wake-dependent way; they are hyperactive during wake, slow-firing during non-REM sleep, and completely quiescent during REM sleep (Mitchell & Weinshenker, 2010). Findings from human data imply that NA-ergic signalling could be the appropriate focus for the creation of therapeutic approaches intended to lessen arousal and stress reactivity (Pitman et al., 2012b).

Serotonin (or 5-HT), which acts as both a neurotransmitter and a neurohormone, aids in the regulation of normal and pathological activity in key brain regions involved in fear training, fear extinction, and coping (Vitalis & Verney, 2022). Owing to its interconnectedness and extensive homeostatic role, lower levels of 5-HT transmission might contribute to PTSD symptoms such as hypervigilance, heightened startle, impulsivity, and intrusive memories (Vermetten & Bremner, 2002). An altered 5-HT signaling in PTSD may affect REMS in several ways. Low serotonin levels can lead to vivid and emotionally strong dreams during REMS as

well as contribute to hypervigilance. Depending on the stressor strength, brain location, and receptor type, 5-HT dysregulation may either lead to abnormally heightened REMS activity or as reported by other studies, may reduce the time spent in REMS (Iwasaki et al., 2018) and weaken the transition from NREMS to REMS, thus leading to severity in PTSD symptomology (Dauvilliers et al., 2015). Selective Serotonin-Reuptake Inhibitors (SSRIs), are antidepressant drugs that are routinely administered (Morigaki et al., 2016). They act by boosting serotonin levels in the brain. They can assist PTSD patients manage their mood, reduce anxiety, and enhance their sleep quality.

There is evidence that stressors trigger mesolimbic **Dopamine** (DA) release, which may influence HPA axis responses in people with PTSD (Sherin & Nemeroff, 2011c). It is unknown whether DA metabolism is always affected in PTSD, although a genetic component associated with DA metabolism, the SLC6A3 (solute carrier family 6 member 3) locus 9R allele, has been reported to govern the development and symptomology of PTSD (Stevens & Jovanovic, 2019). Hypodopaminergic function in the brain's reward circuitry due to gene polymorphism is also known to exacerbate substance use disorders in patients with PTSD (Aliev et al., 2020). Heightened DA levels may be associated with REMS disruption and the subsequent hyperarousal symptoms, though there's a lot of ongoing research to provide a plausible explanation (Torrissi et al., 2019). Furthermore, DA's interactions with other neurotransmitters such as NA and serotonin to collectively induce REMS disruptions is also under investigation.

Γ -Aminobutyric acid (GABA) is a primary inhibitory neurotransmitter in the brain. GABA has powerful anxiolytic properties and dampens behavioural and physiological reactions to stressors, in part by blocking CRH/NA circuits implicated in fear and stress response mediation (Aliev et al., 2020). Experiments have revealed that one or more polymorphisms in the $\alpha 2$ subunit of the GABA receptor, GABRA2, contribute to the development of PTSD in

people who have been exposed to childhood trauma, possibly by amplifying the post-event anxiogenic response and aiding traumatic memory consolidation (Nelson et al., 2009).

GABAergic neurotransmission dysfunction, such as reduced GABA levels or altered GABA receptor function, might potentially contribute to the REMS anomalies seen in PTSD. Increased arousal in PTSD, particularly increased GABAergic activity, may influence the transition to REMS, resulting in shortened REMS duration or increased REMS density (Vanini et al., 2011) (R. Mehta, Singh, Khanday, et al., 2017).

Glutamate is the primary excitatory neurotransmitter in the brain, released by the activation of glucocorticoids, and is subsequently bound to NMDA receptors, which play a role in synaptic plasticity, thereby contributing to the consolidation of traumatic memories. However, the overexposure of neurons to glutamate is known to be excitotoxic, disturbing neuronal integrity in the hippocampus and prefrontal cortex of patients with PTSD (Averill et al., 2017). Increased glutamate levels in certain brain areas, such as the ACC and amygdala, have been postulated to contribute to hyperarousal and greater fear response in PTSD patients, triggering REMS anomalies. Changes in glutamate-GABA balance can interfere with the regular functioning of inhibitory interneurons, which govern REMS-related activities such as muscular paralysis (Mondino et al., 2021b).

The ability of neuropeptides like **Neuropeptide Y** to impair CRH secretion, thereby mitigating EPSPs (excitatory post-synaptic potentials) via activation of pre-synaptic Y2 receptors and inhibition of NA circuits involved in stress and fear responses, is linked to their protective properties (Sabban et al., 2016). Rasmusson et al. (2000), for example, discovered reduced NPY levels in the cerebral fluid of combat veterans with PTSD compared to soldiers without PTSD. Furthermore, Morgan et al. (2000) discovered lower NPY levels in the plasma of female sexual assault victims suffering from PTSD. Lower NPY levels were linked to worsening PTSD symptoms (Sah & Geraciotti, 2013). Recent circuit-level research supported this by

demonstrating that stress recruits peri-LC NPY-expressing neurons that suppress LC NA-ergic neuronal excitability through Y1 receptor-mediated inhibition (Riga et al., 2025). In the context of PTSD-associated REM sleep disturbance, impaired peri-LC NPY signalling may suppress endogenous inhibition of LC activity, thereby permitting sustained noradrenergic hyperactivity which leads to heightened arousal, REMS fragmentation, and persistence of trauma-related nightmares. As aforementioned in this, the lower level of NPY is directly associated with increase of NA levels which in turn is associated with the sleep disruptions. It has been proposed that changes in NPY levels or NPY receptor function in specific brain regions, such as the amygdala and hypothalamus, may influence the stress response and sleep-wake regulation, potentially leading to REMS disturbances in PTSD (Kienast et al., 2019) (Shen et al., 2022).

Factors influencing PTSD-induced variability in REMS

The acquisition of PTSD after a traumatic exposure may involve certain factors potentially affecting how the brain could respond to traumatic stressors and the consequent alterations in an individual's ability to cope with it. This section will go through some of the elements that can influence the propensity to PTSD. We shall concentrate on the effects of ageing, gender differences, and gene-environment interactions.

A Focus on Ageing

Progression into older age is typically characterized by distinctive changes in sleep patterns over which NA-ergic systems have an important control. REMS reduces with the progression of age. Though reduced NA has also been observed in some studies, the consensus view is that resting sympathetic nervous system activity increases progressively with ageing (Esler et al., 1995). Thus, as people age, their plasma NA levels generally rise, which may lead to sympathetic hyperactivity (J. Lee & Kim, 2022). Researchers have concluded ageing-associated symptoms could be due to REMS loss and consequent increase in NA in the brain. Increase in NA levels has also been attributed to pathological causes, for instance, use of

certain prescribed medicines such as SSRIs and NA reuptake inhibitors (Huertas-Ceballos et al., 2008) or, due to certain, neurological disorders like Parkinson's disease, where there is a loss in DA-producing neurons and a consequent increased compensatory NA level (Delaville et al., 2011a). Furthermore, cardiovascular diseases and chronic pain may also result in excessive NA release in older adults (Sealey & Laragh, 2009).

This abnormally increased NA release in elderly persons can increase alpha-sympathetic activity, making the transition to REMS more challenging (R. Mehta et al., 2019; Spindler et al., 2023) which may even predispose the geriatric population to acquiring PTSD post-traumatic exposure (McCarthy, 2021). Moreover, the effectiveness of prazosin, an α -1 adrenoreceptor antagonist, in alleviating severity of PTSD symptomology, (Khaw & Argo, 2020) like REMS associated-nightmares, may also indicate the increased NA levels as the underlying culprit eliciting REMS-linked PTSD outcomes. Although previous epidemiological studies have reported lower rates of PTSD in older adults, scarcity of experiments specifically with older adults leads to a call for research in this domain (Pless Kaiser et al., 2019). As a result, existing estimates of the prevalence of PTSD in older persons may be understated. Studies addressing developmental dynamics between age-dependent physiological changes and PTSD symptoms are required. Maturational gains are also important to consider. For instance, although older age is associated with improved emotional control, emotional well-being, and positivity bias, it is rarely examined how age-related benefits intersect with PTSD psychopathology in later life.

Of note, age-related declines in cognitive and physical functions may also influence coping with REMS symptom manifestations in PTSD. Furthermore, comorbidities with medical, psychiatric, and cognitive problems in older adults and consequent overlapping of sleep and neurochemical changes due to them may make it complicated to study, screen and strategize treatments for this population. For instance, high comorbidities of cardiovascular diseases like

hypertension which records higher amounts of NA in the elderly may tangle with the pathology elicited by a traumatic event. In addition, high prevalence rates of sleep breathing disorders, anxiety disorders, stroke, substance use disorders in the elderly, all reporting SDs and NA levels abnormalities, may put these individuals at a risk of developing PTSD (Thomas et al., 2017). Overall, there is much to be learned about the constructs of PTSD in later life, as well as its assessments, diagnosis, and treatments.

Gender-specific variation

Women are more likely than men to develop chronic PTSD, along with distinct symptom profiles and comorbidities, regardless of the type or frequency of the trauma (Kobayashi & Mellman, 2012a). Men with PTSD often exhibit externalizing symptoms such as aggression and risk-taking behaviours, while women with PTSD tend to exhibit internalizing symptoms including sadness, loss of enjoyment, and social difficulties (Pooley et al., 2018a). These gender differences in symptomatology suggest underlying neurobiological variations in the traumatic stress response (Pooley et al., 2018b). Research indicates that PTSD is associated with gender disparities in sleep problems following trauma. Studies have found that women, compared to men, experience slower waves and consolidated sleep based on polysomnography (PSG) studies in the general population. In both sexes, the development of PTSD has been associated with shorter duration and increased frequency of REM segments, suggesting a potential contribution of REM sleep disruption to the development of PTSD (Mellman et al., 2002). However, disrupted maintenance of sleep was only observed in women, indicating that this sleep disruption following trauma may contribute to women's higher risk of developing PTSD (Kobayashi & Mellman, 2012b).

Neurobiologically, PTSD has been associated with amygdala hyperactivity, hypoactivity in the vmPFC, and decreased functional connectivity between the vmPFC and amygdala (Olf, 2017). In females, the activity in the vmPFC and amygdala may serve as

markers of trauma pathology, including symptoms of anhedonia, whereas male responses to trauma are often unaffected in these brain regions. The data, however, show that vmPFC and amygdala activity may be pertinent markers of trauma pathology for females, but typically not for males, and may be linked to a behavioural trait known as anhedonia in females (Pooley et al., 2018a).

Susceptibility of women to have hyperarousal linked disorders is attributed to gender variations in the brain's arousal centres, specifically LC, which releases NA into forebrain regions to regulate arousal levels, is a potential site for such sex differences. Studies prove that depressed patients' Corticotropin-releasing factor hypersecretion targets the LC in particular (Bangasser et al., 2016a). These findings have led to basic research that revealed enhanced female LC neuronal sensitivity to Corticotropin-releasing factor and the processes underpinning this effect, in addition to higher prevalence of stress-related psychiatric illnesses in women. Presynaptic modulation of NA release, which can happen through oestrogen regulation of NA synthesis enzymes, has been suggested to be a mechanism by which ovarian hormones boost NA levels in LC target areas. Gonadal hormone modulation of NA-degrading enzymes is another mechanism that could contribute to greater amounts of NA in LC target areas. Catechol-O-methyltransferase (COMT) is the subject of this study because single nucleotide polymorphisms (SNPs) on the COMT gene have been linked to a variety of psychiatric diseases that differ between the sexes (Bangasser et al., 2016b). Besides, the imbalance between NA release and postsynaptic adrenergic receptors is likely to contribute to hyperarousal symptoms in stress-related psychiatric disorders, including PTSD. This excess NA may increase the risk of hyperarousal symptoms in women, particularly during their reproductive years when estrogen levels fluctuate frequently. Notably, the rates of depression in females are higher than in males before menopause and post-puberty, indicating a potential hormonal influence on psychiatric disorders.

Gene-environment interactions

When the effect of genotype on risk for a condition alters depending on the presence or absence of an environmental pathogen, this is referred to as a gene-environment interaction (Koenen et al., 2008). PTSD is a relevant topic for G×E studies as one of the criteria for diagnosis of the disorder is pre-exposure to the traumatic event such as maltreated children, female victims of sexual abuse, male and female combatants in war, etc (Cornelis et al., 2010; D. Mehta & Binder, 2012). Research suggested that the adult generation of Holocaust survivors with PTSD had a higher risk of PTSD following trauma than adult children of Holocaust survivors without PTSD, clearly indicating that relatives of probands with PTSD are more prone to this disorder (Yehuda et al., 2001). Individual differences in personality are most likely to blame for gene-environmental connections. For example, research has discovered that childhood externalising relates to both the risk of trauma exposure and the development of PTSD in adulthood (KOENEN et al., 2007).

Epigenetic processes control gene expression without changing the DNA sequence. Epigenetic modifications can be accumulated throughout life and can moderate the impact of the environment on gene expression. COMT is an enzyme that catalyses and inactivates catecholamines such as DA, NA, and adrenaline (Boscarino et al., 2012; Kolassa et al., 2010; Valente et al., 2011) and COMT Val158Met polymorphism has been linked to PTSD development as well as elimination of the conditioned fear response. Previously, this polymorphism was the subject of research that related it to anxiety problems (Olsson et al., 2005). A recent study linked Val158Met to a slower recovery from stressful events or painful conditions, a trait known as resilience (Guillén-Burgos & Gutiérrez-Ruiz, 2018). The sleep related genes were found to be impacted in chronic blast-exposed military personnel including methylation of NTSR1 and SPON1 and differentially methylated regions for CCDC68 and COMT (associated with sleep difficulties experienced by the individuals) (Raza et al., 2023).

A large proportion of male patients with current PTSD gathered from post-war had higher monoamine oxidase A (MAOA) gene DNA methylation, which could be an epigenetic marker of disease state and severity. This was especially true for PTSD symptom clusters linked with reliving trauma and hyperarousal, which have been postulated to be primarily controlled by the NA-ergic system, whose hyperactivity causes REMS disruption. Gene hypermethylation results in reduced expression (Raza et al., 2023), in this case MAOA hypermethylation may result in decreased MAOA transcription and, as a result, enhanced NA-ergic signalling. The current findings aid in directing future tailored therapy approaches in male PTSD patients with elevated MAOA methylation, showing a beneficial usage of anti-adrenergic drugs, particularly for re-experiencing and hyperarousal symptoms (Ziegler et al., 2018).

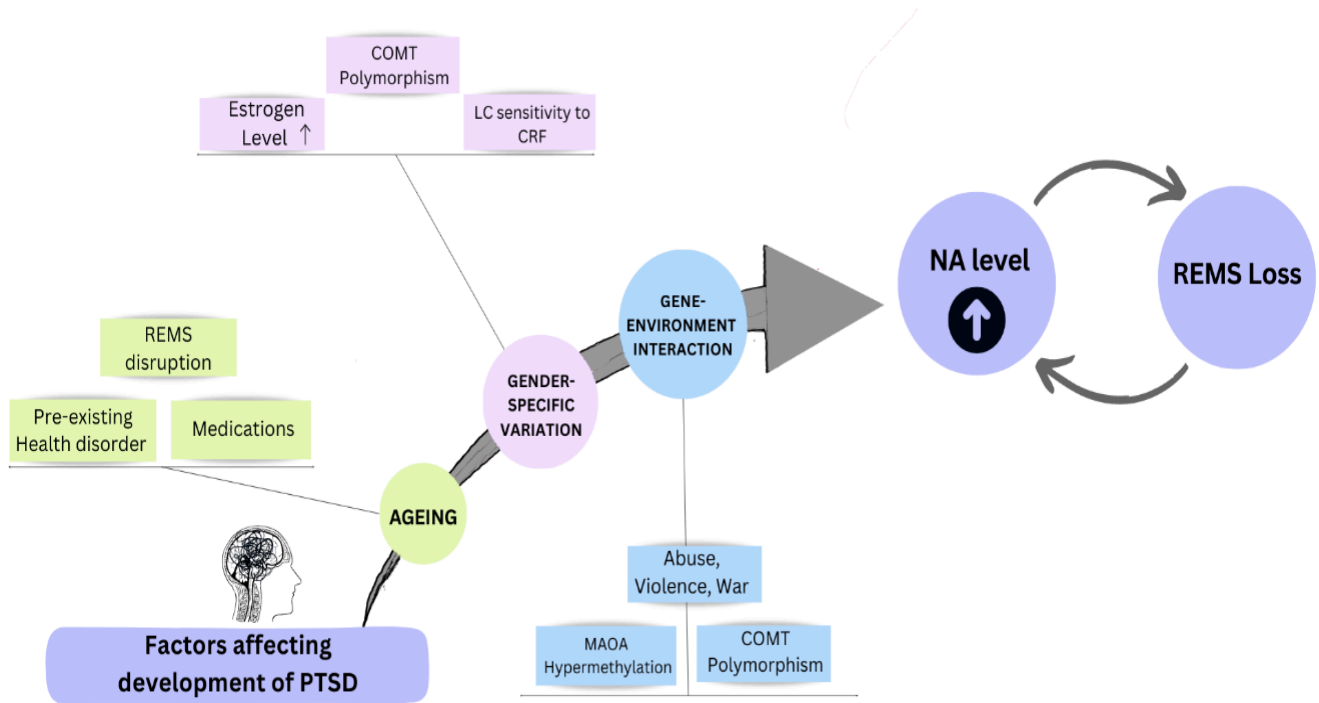


Figure 3.

Factors affecting the development of PTSD: NA levels may be abnormally influenced by certain factors, possibly resulting in a proneness to acquire PTSD following a traumatic event. (1) Older adults may be prone to developing PTSD owing to weakened coping systems (R. Mehta et al., 2019). (2) Women are more likely to have PTSD than men due to variance in biological factors (Bangasser et al., 2016b). (3) Certain epigenetics may ensure susceptibility to developing PTSD (Guillén-Burgos & Gutiérrez-Ruiz, 2018; Ziegler et al., 2018).

Amelioration

The studies addressed in this review imply that NA-ergic system is a critical, changeable link in an interconnected network of stress-response system, involving REMS regulation and hence, medicines targeting the central NA system may have a therapeutic benefit in the treatment of this disorder. Overall, sleep issues have an impact on the development and intensity of PTSD symptoms, as well as patients' quality of life and functioning. After decades of research and clinical trials, debilitation of pathological NA-ergic hyperactivity has emerged as a crucial therapeutic strategy to treat REMS aberrancies in PTSD.

NA-reducing strategies

In terms of treatments addressing levels of NA, Prazosin, a centrally acting alpha 1-adrenoceptor antagonist, has been shown in small studies of PTSD patients, to prevent or greatly decrease the expression of many but not all of the symptoms of PTSD, significantly reducing nightmares and insomnia (Hendrickson & Raskind, 2016b; Khazaie et al., 2016). Prazosin emerged as the main agent to treat sleep problems in PTSD, by decreasing the NA effects at brain α -1 adrenoreceptors, such as in that of basolateral amygdala, hippocampus and PFC (Berardis et al., 2015; Ketenci et al., 2020; Paiva et al., 2021). While beta-blockers like propranolol are typically used to treat cardiovascular issues, they may also aid with hyperarousal and anxiety symptoms linked with PTSD. Propranolol appears to block the beta-adrenergic neurotransmission in the basolateral amygdala, thereby inducing NA enhance-conditioning (Mallet et al., 2022; McGaugh, 2002).

Animal studies have shown that alpha-2A receptor stimulation can protect PFC function in patients with PTSD, with simultaneously weakening amygdala, and regulating NA cell firing (Arnsten et al., 2015). Thus, stimulating alpha-2A receptors with guanfacine or clonidine can be useful in reducing the symptoms of PTSD. Open label trials indicate that guanfacine may be especially beneficial in treating traumatised children and adolescents

(Connor et al., 2013; Fox & Sinha, 2014). Of note, in reaction to distressing experiences, NPY can operate as a stress buffer by lowering NA hyperactivity. Dysfunction in peri-NPY/Y1 circuit can result in LC hyperactivity with excessive NA-ergic firing and REM sleep fragmentation observed in PTSD. Consequently, selective enhancement of NPY-Y1 receptor signalling can be a promising therapeutic strategy to alleviate REM disruption and hyperarousal symptoms in PTSD (Riga et al., 2025). Additionally, neuropeptide agonists may be useful in the treatment of sleep problems and hyperarousal states (Shen et al., 2022). However, an in-depth understanding of this class is being pursued actively (Tasan et al., 2016).

Strategies targeting NA reduction in combination with other neurotransmitters

Over the decades, antidepressants like SSRIs have been used to treat PTSD because of the high prevalence of comorbid depressive symptoms (Maher et al., 2006). However, their adverse effects on REMS duration and latency have been identified lately (Hutka et al., 2021). In both patients and healthy volunteers, most antidepressants like the serotonin and NA reuptake inhibitors (SNRI), NA reuptake inhibitors (NARI), monoamine oxidase inhibitors (MAOI); the action of MAOI works though by preventing the breakdown of monoamines by enzyme Monoamine Oxidase have been found to suppress REMS (Wichniak et al., 2017).

In contrast, tetracyclic antidepressants like mianserin and mirtazapine having a combined serotonergic-NA-ergic effect, appear to have a positive influence on sleep. They function as antagonists at the 5-HT₂ and H₁ receptors (unlike reuptake inhibitors) with both resulting in increased REMS duration and improving overall sleep efficiency (Mendlewicz et al., 1985) (Schittecatte et al., 2002). Ongoing research shows that use of this class might be superior to the traditional use of SSRIs and SNRIs to treat important PTSD symptoms.

While numerous studies have been conducted and further being explored to address NA levels for REMS disruption treatment, other neurochemicals have been paving novel pharmacotherapies, independently or in combinations. Use of hypnotics like zolpidem and

zaleplon that target GABA receptors has suggested greater sleep-inducing abilities and fewer side effects (as opposed to the traditional use and abuse of benzodiazepines) (Pagel & Parnes, 2001). Furthermore, anticonvulsants, some of which have GABA like properties, such as pregabalin, gabapentin, tiagabine, lamotrigine, and topiramate have been researched and explored (Garakani et al., 2020).

Conclusions and Future Directions

PTSD is typically accompanied with various and diverse sleep disturbances that impact its onset, maintenance, and treatment; research convincingly suggests. Thus, appropriate addressal of sleep, particularly REMS complications, is among one of the most crucial aspects of treating PTSD. Increased levels of NA observed in PTSD may play an important role in inducing/worsening post-traumatic REMS abnormalities, thus making it a crucial therapeutic target to treat PTSD symptomology by resolving REMS disturbances. Moreover, the risk of developing PTSD in the aftermath of severe trauma is also determined by interplay of multiple factors such as ageing, gender difference and gene-environment interactions, owing to NA-mediated REMS quality.

In this review, we propose that advances in the neuroscience of sleep may help to guide assessment and developing novel interventions to PTSD treatment. In addition to NA-specific therapeutics, NPY-enhancing as well as combinations with other neurotransmitter-targeted strategies could be deployed in individual cases by the clinicians. Finally, to integrate generic and population-specific elements, research must be expanded to encompass larger and more diverse populations of those affected by PTSD.

References

- Aliev, G., Beeraka, N. M., Nikolenko, V. N., Svistunov, A. A., Rozhnova, T., Kostyuk, S., Cherkesov, I., Gavryushova, L. V., Chekhonatsky, A. A., Mikhaleva, L. M., Somasundaram, S. G., Avila-Rodriguez, M. F., & Kirkland, C. E. (2020). Neurophysiology and Psychopathology Underlying PTSD and Recent Insights into the PTSD Therapies—A Comprehensive Review. *Journal of Clinical Medicine*, 9(9), 2951. <https://doi.org/10.3390/jcm9092951>
- Arnsten, A. F. T., Raskind, M. A., Taylor, F. B., & Connor, D. F. (2015). The effects of stress exposure on prefrontal cortex: Translating basic research into successful treatments for post-traumatic stress disorder. *Neurobiology of Stress*, 1, 89–99. <https://doi.org/10.1016/j.ynstr.2014.10.002>
- Aston-Jones, G., & Bloom, F. E. (1981). Activity of norepinephrine-containing locus coeruleus neurons in behaving rats anticipates fluctuations in the sleep-waking cycle. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 1(8), 876–886. <https://doi.org/10.1523/JNEUROSCI.01-08-00876.1981>
- Averill, L. A., Purohit, P., Averill, C. L., Boesl, M. A., Krystal, J. H., & Abdallah, C. G. (2017). Glutamate dysregulation and glutamatergic therapeutics for PTSD: Evidence from human studies. *Neuroscience Letters*, 649, 147–155. <https://doi.org/10.1016/j.neulet.2016.11.064>
- Bangasser, D. A., Wiersielis, K. R., & Khantsis, S. (2016a). Sex Differences in the Locus Coeruleus-Norepinephrine System and its Regulation by Stress. *Brain Research*, 1641(Pt B), 177–188. <https://doi.org/10.1016/j.brainres.2015.11.021>

- Bangasser, D. A., Wiersielis, K. R., & Khantsis, S. (2016b). Sex Differences in the Locus Coeruleus-Norepinephrine System and its Regulation by Stress. *Brain Research*, 1641(Pt B), 177–188. <https://doi.org/10.1016/j.brainres.2015.11.021>
- Belleville, G., Guay, S., & Marchand, A. (2011). Persistence of sleep disturbances following cognitive-behavior therapy for posttraumatic stress disorder. *Journal of Psychosomatic Research*, 70(4), 318–327. <https://doi.org/10.1016/j.jpsychores.2010.09.022>
- Berardis, D., Marini, S., Serroni, N., Iasevoli, F., Tomasetti, C., Bartolomeis, A., Mazza, M., Tempesta, D., Valchera, A., Fornaro, M., Pompili, M., Sepede, G., Vellante, F., Orsolini, L., Martinotti, G., & Giannantonio, M. (2015). Targeting the Noradrenergic System in Posttraumatic Stress Disorder: A Systematic Review and Meta-Analysis of Prazosin Trials. *Current Drug Targets*, 16(10), 1094–1106. <https://doi.org/10.2174/1389450116666150506114108>
- Blanchard, E. B., Kolb, L. C., Prins, A., Gates, S., & McCoy, G. C. (1991a). Changes in plasma norepinephrine to combat-related stimuli among Vietnam veterans with posttraumatic stress disorder. *The Journal of Nervous and Mental Disease*, 179(6), 371–373. <https://doi.org/10.1097/00005053-199106000-00012>
- Blanchard, E. B., Kolb, L. C., Prins, A., Gates, S., & McCoy, G. C. (1991b). Changes in plasma norepinephrine to combat-related stimuli among Vietnam veterans with posttraumatic stress disorder. *The Journal of Nervous and Mental Disease*, 179(6), 371–373. <https://doi.org/10.1097/00005053-199106000-00012>
- Boscarino, J. A., Erlich, P. M., Hoffman, S. N., & Zhang, X. (2012). Higher *FKBP5*, *COMT*, *CHRNA5*, and *CRHR1* allele burdens are associated with PTSD and interact with trauma

exposure: Implications for neuropsychiatric research and treatment. *Neuropsychiatric Disease and Treatment*, 8, 131–139. <https://doi.org/10.2147/NDT.S29508>

- Bowirrat, A., J.H. Chen, T., Blum, K., Madigan, M., A. Bailey, J., Lih Chuan Chen, A., William Downs, B., R. Braverman, E., Radi, S., L. Waite, R., Kerner, M., Giordano, J., Morse, S., Oscar-Berman, M., & Gold, M. (2010). Neuro-psychopharmacogenetics and Neurological Antecedents of Posttraumatic Stress Disorder: Unlocking the Mysteries of Resilience and Vulnerability. *Current Neuropsychopharmacology*, 8(4), 335–358. <https://doi.org/10.2174/157015910793358123>
- Bremner, J. D., Randall, P., Scott, T. M., Bronen, R. A., Seibyl, J. P., Southwick, S. M., Delaney, R. C., McCarthy, G., Charney, D. S., & Innis, R. B. (1995). MRI-based measurement of hippocampal volume in patients with combat-related posttraumatic stress disorder. *The American Journal of Psychiatry*, 152(7), 973–981. <https://doi.org/10.1176/ajp.152.7.973>
- Bremner, J. D., & Wittbrodt, M. T. (2020). Stress, the brain, and trauma spectrum disorders. In *International Review of Neurobiology* (Vol. 152, pp. 1–22). Elsevier. <https://doi.org/10.1016/bs.irn.2020.01.004>
- Brownlow, J. A., McLean, C. P., Gehrman, P. R., Harb, G. C., Ross, R. J., & Foa, E. B. (2016). Influence of Sleep Disturbance on Global Functioning After Posttraumatic Stress Disorder Treatment: PTSD Treatment, Sleep, Global Functioning. *Journal of Traumatic Stress*, 29(6), 515–521. <https://doi.org/10.1002/jts.22139>
- Brun, J., Chamba, G., Khalfallah, Y., Girard, P., Boissy, I., Bastuji, H., Sassolas, G., & Claustrat, B. (1998). Effect of modafinil on plasma melatonin, cortisol and growth hormone rhythms, rectal temperature and performance in healthy subjects during a 36

- h sleep deprivation. *Journal of Sleep Research*, 7(2), 105–114.
<https://doi.org/10.1046/j.1365-2869.1998.00100.x>
- Bryant, R. A., Creamer, M., O'Donnell, M., Silove, D., & McFarlane, A. C. (2010). Sleep Disturbance Immediately Prior to Trauma Predicts Subsequent Psychiatric Disorder. *Sleep*, 33(1), 69–74. <https://doi.org/10.1093/sleep/33.1.69>
- Carter, M. E., Yizhar, O., Chikahisa, S., Nguyen, H., Adamantidis, A., Nishino, S., Deisseroth, K., & de Lecea, L. (2010). Tuning arousal with optogenetic modulation of locus coeruleus neurons. *Nature Neuroscience*, 13(12), 1526–1533.
<https://doi.org/10.1038/nn.2682>
- Christiansen, D. M., & Hansen, M. (2015). Accounting for sex differences in PTSD: A multi-variable mediation model. *European Journal of Psychotraumatology*, 6, 26068.
<https://doi.org/10.3402/ejpt.v6.26068>
- Chrousos, G. P., & Gold, P. W. (1992). The concepts of stress and stress system disorders. Overview of physical and behavioral homeostasis. *JAMA*, 267(9), 1244–1252.
- Colvonen, P. J., Straus, L. D., Acheson, D., & Gehrman, P. (2019). A Review of the Relationship Between Emotional Learning and Memory, Sleep, and PTSD. *Current Psychiatry Reports*, 21(1), 2. <https://doi.org/10.1007/s11920-019-0987-2>
- Connor, D. F., Grasso, D. J., Slivinsky, M. D., Pearson, G. S., & Banga, A. (2013). An Open-Label Study of Guanfacine Extended Release for Traumatic Stress Related Symptoms in Children and Adolescents. *Journal of Child and Adolescent Psychopharmacology*, 23(4), 244–251. <https://doi.org/10.1089/cap.2012.0119>
- Cornelis, M. C., Nugent, N. R., Amstadter, A. B., & Koenen, K. C. (2010). Genetics of Post-Traumatic Stress Disorder: Review and Recommendations for Genome-Wide

Association Studies. *Current Psychiatry Reports*, 12(4), 313–326.

<https://doi.org/10.1007/s11920-010-0126-6>

Dauvilliers, Y., Tafti, M., & Landolt, H. P. (2015). Catechol-O-methyltransferase, dopamine, and sleep-wake regulation. *Sleep Medicine Reviews*, 22, 47–53.

<https://doi.org/10.1016/j.smr.2014.10.006>

Delaville, C., Deurwaerdère, P. D., & Benazzouz, A. (2011a). Noradrenaline and Parkinson's Disease. *Frontiers in Systems Neuroscience*, 5.

<https://doi.org/10.3389/fnsys.2011.00031>

Esler, M. D., Thompson, J. M., Kaye, D. M., Turner, A. G., Jennings, G. L., Cox, H. S., Lambert, G. W., & Seals, D. R. (1995). Effects of Aging on the Responsiveness of the Human Cardiac Sympathetic Nerves to Stressors. *Circulation*, 91(2), 351–358.

<https://doi.org/10.1161/01.CIR.91.2.351>

Fox, H., & Sinha, R. (2014). The Role of Guanfacine as a Therapeutic Agent to Address Stress-Related Pathophysiology in Cocaine-Dependent Individuals. In *Advances in Pharmacology* (Vol. 69, pp. 217–265). Elsevier.

<https://doi.org/10.1016/B978-0-12-420118-7.00006-8>

Fraigne, J. J., Torontali, Z. A., Snow, M. B., & Peever, J. H. (2015). REM Sleep at its Core—Circuits, Neurotransmitters, and Pathophysiology. *Frontiers in Neurology*, 6,

123. <https://doi.org/10.3389/fneur.2015.00123>

Galovski, T. E., Harik, J. M., Blain, L. M., Elwood, L., Gloth, C., & Fletcher, T. D. (2016). Augmenting cognitive processing therapy to improve sleep impairment in PTSD: A randomized controlled trial. *Journal of Consulting and Clinical Psychology*, 84(2),

167–177. <https://doi.org/10.1037/ccp0000059>

- Garakani, A., Murrough, J. W., Freire, R. C., Thom, R. P., Larkin, K., Buono, F. D., & Iosifescu, D. V. (2020). Pharmacotherapy of Anxiety Disorders: Current and Emerging Treatment Options. *Frontiers in Psychiatry, 11*, 595584. <https://doi.org/10.3389/fpsy.2020.595584>
- Garthus-Niegel, S., Ayers, S., Von Soest, T., Torgersen, L., & Eberhard-Gran, M. (2015). Maintaining factors of posttraumatic stress symptoms following childbirth: A population-based, two-year follow-up study. *Journal of Affective Disorders, 172*, 146–152. <https://doi.org/10.1016/j.jad.2014.10.003>
- Germain, A., Buysse, D. J., & Nofzinger, E. (2008). Sleep-specific mechanisms underlying posttraumatic stress disorder: Integrative review and neurobiological hypotheses. *Sleep Medicine Reviews, 12*(3), 185–195. <https://doi.org/10.1016/j.smrv.2007.09.003>
- Gillin, J. C., Jacobs, L. S., Fram, D. H., & Snyder, F. (1972). Acute effect of a glucocorticoid on normal human sleep. *Nature, 237*(5355), 398–399. <https://doi.org/10.1038/237398a0>
- Godoy, L. D., Rossignoli, M. T., Delfino-Pereira, P., Garcia-Cairasco, N., & de Lima Umeoka, E. H. (2018). A Comprehensive Overview on Stress Neurobiology: Basic Concepts and Clinical Implications. *Frontiers in Behavioral Neuroscience, 12*. <https://www.frontiersin.org/articles/10.3389/fnbeh.2018.00127>
- Gradus, J. L., & Galea, S. (2023). Moving from traumatic events to traumatic experiences in the study of traumatic psychopathology. *American Journal of Epidemiology*, kwad126. <https://doi.org/10.1093/aje/kwad126>
- Guillén-Burgos, H. F., & Gutiérrez-Ruiz, K. (2018). Genetic advances in post-traumatic stress disorder. *Revista Colombiana de Psiquiatría (English Ed.)*, 47(2), 108–118. <https://doi.org/10.1016/j.rcpeng.2018.03.001>

- Hendrickson, R. C., & Raskind, M. A. (2016b). Noradrenergic dysregulation in the pathophysiology of PTSD. *Experimental Neurology*, 284(Pt B), 181–195.
<https://doi.org/10.1016/j.expneurol.2016.05.014>
- Hendrickson, R. C., Raskind, M. A., Millard, S. P., Sikkema, C., Terry, G. E., Pagulayan, K. F., Li, G., & Peskind, E. R. (2018). Evidence for altered brain reactivity to norepinephrine in Veterans with a history of traumatic stress. *Neurobiology of Stress*, 8, 103–111. <https://doi.org/10.1016/j.ynstr.2018.03.001>
- Huertas-Ceballos, A. A., Logan, S., Bennett, C., & Macarthur, C. (2008). Psychosocial interventions for recurrent abdominal pain (RAP) and irritable bowel syndrome (IBS) in childhood. In The Cochrane Collaboration (Ed.), *Cochrane Database of Systematic Reviews* (p. CD003014.pub2). John Wiley & Sons, Ltd.
<https://doi.org/10.1002/14651858.CD003014.pub2>
- Hutka, P., Krivosova, M., Muchova, Z., Tonhajzerova, I., Hamrakova, A., Mlyncekova, Z., Mokry, J., & Ondrejka, I. (2021). Association of Sleep Architecture and Physiology with Depressive Disorder and Antidepressants Treatment. *International Journal of Molecular Sciences*, 22(3), 1333. <https://doi.org/10.3390/ijms22031333>
- Illes, P., Finta, E. P., & Nieber, K. (1993). Neuropeptide Y potentiates via Y2-receptors the inhibitory effect of noradrenaline in rat locus coeruleus neurones. *Naunyn-Schmiedeberg's Archives of Pharmacology*, 348(5), 546–548.
<https://doi.org/10.1007/BF00173217>
- Iwasaki, K., Komiya, H., Kakizaki, M., Miyoshi, C., Abe, M., Sakimura, K., Funato, H., & Yanagisawa, M. (2018). Ablation of Central Serotonergic Neurons Decreased REM Sleep and Attenuated Arousal Response. *Frontiers in Neuroscience*, 12, 535.
<https://doi.org/10.3389/fnins.2018.00535>

- Kalmbach, D. A., Cuamatzi-Castelan, A. S., Tonnu, C. V., Tran, K. M., Anderson, J. R., Roth, T., & Drake, C. L. (2018). Hyperarousal and sleep reactivity in insomnia: Current insights. *Nature and Science of Sleep, 10*, 193–201.
<https://doi.org/10.2147/NSS.S138823>
- Ketenci, S., Acet, N. G., Sarıdoğan, G. E., Aydın, B., Cabadak, H., & Gören, M. Z. (2020). The Neurochemical Effects of Prazosin Treatment on Fear Circuitry in a Rat Traumatic Stress Model. *Clinical Psychopharmacology and Neuroscience, 18*(2), 219–230. <https://doi.org/10.9758/cpn.2020.18.2.219>
- Khaw, C., & Argo, T. (2020). Prazosin Outcomes in Older Veterans With Posttraumatic Stress Disorder. *Federal Practitioner: For the Health Care Professionals of the VA, DoD, and PHS, 37*(2), 72–78.
- Khazaie, H., Nasouri, M., & Ghadami, M. R. (2016). Prazosin for Trauma Nightmares and Sleep Disturbances in Combat Veterans with Post-Traumatic Stress Disorder. *Iranian Journal of Psychiatry and Behavioral Sciences, 10*(3).
<https://doi.org/10.17795/ijpbs-2603>
- Kienast, C., Gunga, H.-C., & Steinach, M. (2019). Neuropeptide Y – Its role in human performance and extreme environments. *REACH, 14–15*, 100032.
<https://doi.org/10.1016/j.reach.2019.100032>
- Kim, Y., Chen, L., McCarley, R. W., & Strecker, R. E. (2013). Sleep allostasis in chronic sleep restriction: The role of the norepinephrine system. *Brain Research, 1531*, 9–16.
<https://doi.org/10.1016/j.brainres.2013.07.048>
- Kjaerby, C., Andersen, M., Hauglund, N., Ding, F., Wang, W., Xu, Q., Deng, S., Kang, N., Peng, S., Sun, Q., Dall, C., Jørgensen, P. K., Feng, J., Li, Y., Weikop, P., Hirase, H., & Nedergaard, M. (2020). *Dynamic fluctuations of the locus coeruleus-norepinephrine*

system underlie sleep state transitions (p. 2020.09.01.274977). bioRxiv.

<https://doi.org/10.1101/2020.09.01.274977>

Kobayashi, I., & Mellman, T. A. (2012a). Gender Differences in Sleep During the Aftermath of Trauma and the Development of Posttraumatic Stress Disorder. *Behavioral Sleep Medicine*, 10(3), 180–190. <https://doi.org/10.1080/15402002.2011.654296>

Kobayashi, I., & Mellman, T. A. (2012b). Gender Differences in Sleep During the Aftermath of Trauma and the Development of Posttraumatic Stress Disorder. *Behavioral Sleep Medicine*, 10(3), 180–190. <https://doi.org/10.1080/15402002.2011.654296>

KOENEN, K. C., MOFFITT, T. E., POULTON, R., MARTIN, J., & CASPI, A. (2007). Early childhood factors associated with the development of post-traumatic stress disorder: Results from a longitudinal birth cohort. *Psychological Medicine*, 37(2), 181–192. <https://doi.org/10.1017/S0033291706009019>

Koenen, K. C., Nugent, N. R., & Amstadter, A. B. (2008). Gene-environment interaction in posttraumatic stress disorder. *European Archives of Psychiatry and Clinical Neuroscience*, 258(2), 82–96. <https://doi.org/10.1007/s00406-007-0787-2>

Koenigs, M., & Grafman, J. (2009). Posttraumatic Stress Disorder: The Role of Medial Prefrontal Cortex and Amygdala. *The Neuroscientist*, 15(5), 540–548. <https://doi.org/10.1177/1073858409333072>

Koffel, E., Polusny, M. A., Arbisi, P. A., & Erbes, C. R. (2013). Pre-deployment daytime and nighttime sleep complaints as predictors of post-deployment PTSD and depression in National Guard troops. *Journal of Anxiety Disorders*, 27(5), 512–519. <https://doi.org/10.1016/j.janxdis.2013.07.003>

Kolassa, I.-T., Kolassa, S., Ertl, V., Papassotiropoulos, A., & De Quervain, D. J.-F. (2010).

The Risk of Posttraumatic Stress Disorder After Trauma Depends on Traumatic Load and the Catechol-O-Methyltransferase Val158Met Polymorphism. *Biological Psychiatry*, 67(4), 304–308. <https://doi.org/10.1016/j.biopsych.2009.10.009>

Kumar, A., Rinwa, P., Kaur, G., & Machawal, L. (2013). Stress: Neurobiology, consequences and management. *Journal of Pharmacy & Bioallied Sciences*, 5(2), 91–97.

<https://doi.org/10.4103/0975-7406.111818>

Kung, S., Espinel, Z., & Lapid, M. I. (2012). Treatment of nightmares with prazosin: A systematic review. *Mayo Clinic Proceedings*, 87(9), 890–900.

<https://doi.org/10.1016/j.mayocp.2012.05.015>

Larsen, S. E., Fleming, C. J. E., & Resick, P. A. (2019). Residual symptoms following empirically supported treatment for PTSD. *Psychological Trauma: Theory, Research, Practice, and Policy*, 11(2), 207–215. <https://doi.org/10.1037/tra0000384>

Lee, H. S., Lee, B. Y., & Waterhouse, B. D. (2005). Retrograde study of projections from the tuberomammillary nucleus to the dorsal raphe and the locus coeruleus in the rat. *Brain Research*, 1043(1–2), 65–75. <https://doi.org/10.1016/j.brainres.2005.02.050>

Lee, J., & Kim, H.-J. (2022). Normal Aging Induces Changes in the Brain and Neurodegeneration Progress: Review of the Structural, Biochemical, Metabolic, Cellular, and Molecular Changes. *Frontiers in Aging Neuroscience*, 14, 931536. <https://doi.org/10.3389/fnagi.2022.931536>

Leproult, R., Copinschi, G., Buxton, O., & Van Cauter, E. (1997). Sleep loss results in an elevation of cortisol levels the next evening. *Sleep*, 20(10), 865–870.

- Liberzon, I., Abelson, J. L., Flagel, S. B., Raz, J., & Young, E. A. (1999a). Neuroendocrine and psychophysiologic responses in PTSD: A symptom provocation study. *Neuropsychopharmacology: Official Publication of the American College of Neuropsychopharmacology*, 21(1), 40–50.
[https://doi.org/10.1016/S0893-133X\(98\)00128-6](https://doi.org/10.1016/S0893-133X(98)00128-6)
- Liberzon, I., Abelson, J. L., Flagel, S. B., Raz, J., & Young, E. A. (1999b). Neuroendocrine and psychophysiologic responses in PTSD: A symptom provocation study. *Neuropsychopharmacology: Official Publication of the American College of Neuropsychopharmacology*, 21(1), 40–50.
[https://doi.org/10.1016/S0893-133X\(98\)00128-6](https://doi.org/10.1016/S0893-133X(98)00128-6)
- Lo Martire, V., Caruso, D., Palagini, L., Zoccoli, G., & Bastianini, S. (2020). Stress & sleep: A relationship lasting a lifetime. *Neuroscience and Biobehavioral Reviews*, 117, 65–77. <https://doi.org/10.1016/j.neubiorev.2019.08.024>
- Lupien, S. J., & McEwen, B. S. (1997). The acute effects of corticosteroids on cognition: Integration of animal and human model studies. *Brain Research. Brain Research Reviews*, 24(1), 1–27. [https://doi.org/10.1016/s0165-0173\(97\)00004-0](https://doi.org/10.1016/s0165-0173(97)00004-0)
- Luppi, P.-H., Gervasoni, D., Verret, L., Goutagny, R., Peyron, C., Salvert, D., Leger, L., & Fort, P. (2006). Paradoxical (REM) sleep genesis: The switch from an aminergic-cholinergic to a GABAergic-glutamatergic hypothesis. *Journal of Physiology, Paris*, 100(5–6), 271–283.
<https://doi.org/10.1016/j.jphysparis.2007.05.006>
- Maher, M. J., Rego, S. A., & Asnis, G. M. (2006). Sleep disturbances in patients with post-traumatic stress disorder: Epidemiology, impact and approaches to management. *CNS Drugs*, 20(7), 567–590. <https://doi.org/10.2165/00023210-200620070-00003>

- Mallet, C., Chick, C. F., Maatoug, R., Fossati, P., Brunet, A., & Millet, B. (2022). Memory reconsolidation impairment using the β -adrenergic receptor blocker propranolol reduces nightmare severity in patients with posttraumatic stress disorder: A preliminary study. *Journal of Clinical Sleep Medicine*, 18(7), 1847–1855.
<https://doi.org/10.5664/jcsm.10010>
- Matosevich, N., & Nir, Y. (2021). Noradrenaline: Sleep on it. *Current Biology*, 31(22), R1477–R1479. <https://doi.org/10.1016/j.cub.2021.10.028>
- McCarthy, C. E. (2021). Sleep Disturbance, Sleep Disorders and Co-Morbidities in the Care of the Older Person. *Medical Sciences*, 9(2), 31.
<https://doi.org/10.3390/medsci9020031>
- McEwen, B. S. (2017). Neurobiological and Systemic Effects of Chronic Stress. *Chronic Stress (Thousand Oaks, Calif.)*, 1, 2470547017692328.
<https://doi.org/10.1177/2470547017692328>
- McGaugh, J. (2002). Amygdala Modulation of Memory Consolidation: Interaction with Other Brain Systems. *Neurobiology of Learning and Memory*, 78(3), 539–552.
<https://doi.org/10.1006/nlme.2002.4082>
- Mehta, D., & Binder, E. B. (2012). Gene $\times$ environment vulnerability factors for PTSD: The HPA-axis. *Neuropharmacology*, 62(2), 654–662.
<https://doi.org/10.1016/j.neuropharm.2011.03.009>
- Mehta, R., Giri, S., & Mallick, B. N. (2020). REM sleep loss–induced elevated noradrenaline could predispose an individual to psychosomatic disorders: A review focused on proposal for prediction, prevention, and personalized treatment. *EPMA Journal*, 11(4), 529–549. <https://doi.org/10.1007/s13167-020-00222-1>

- Mehta, R., Kumar, A., & Mallick, B. N. (2019). Association Between Ageing and REM Sleep Loss: Noradrenaline Acting as a Mediator. In P. C. Rath (Ed.), *Models, Molecules and Mechanisms in Biogerontology* (pp. 109–126). Springer Singapore. https://doi.org/10.1007/978-981-13-3585-3_6
- Mehta, R., Singh, A., Bókkon, I., & Nath Mallick, B. (2016). REM sleep and its Loss-Associated Epigenetic Regulation with Reference to Noradrenaline in Particular. *Current Neuropharmacology*, 14(1), 28–40. <https://doi.org/10.2174/1570159x13666150414185737>
- Mehta, R., Singh, A., & Mallick, B. N. (2017). Disciplined sleep for healthy living: Role of noradrenaline. *World Journal of Neurology*, 7(1), 6. <https://doi.org/10.5316/wjn.v7.i1.6>
- Mehta, R., Singh, S., Khanday, M. A., & Mallick, B. N. (2017). Reciprocal changes in noradrenaline and GABA levels in discrete brain regions upon rapid eye movement sleep deprivation in rats. *Neurochemistry International*, 108, 190–198. <https://doi.org/10.1016/j.neuint.2017.03.016>
- Mellman, T. A., Bustamante, V., Fins, A. I., Pigeon, W. R., & Nolan, B. (2002). REM Sleep and the Early Development of Posttraumatic Stress Disorder. *American Journal of Psychiatry*, 159(10), 1696–1701. <https://doi.org/10.1176/appi.ajp.159.10.1696>
- Mellman, T. A., Kumar, A., Kulick-Bell, R., Kumar, M., & Nolan, B. (1995a). Nocturnal/daytime urine noradrenergic measures and sleep in combat-related PTSD. *Biological Psychiatry*, 38(3), 174–179. [https://doi.org/10.1016/0006-3223\(94\)00238-X](https://doi.org/10.1016/0006-3223(94)00238-X)
- Mellman, T. A., Kumar, A., Kulick-Bell, R., Kumar, M., & Nolan, B. (1995b). Nocturnal/daytime urine noradrenergic measures and sleep in combat-related PTSD.

Biological Psychiatry, 38(3), 174–179.

[https://doi.org/10.1016/0006-3223\(94\)00238-X](https://doi.org/10.1016/0006-3223(94)00238-X)

Mellman, T. A., Kumar, A., Kulick-Bell, R., Kumar, M., & Nolan, B. (1995c).

Nocturnal/daytime urine noradrenergic measures and sleep in combat-related PTSD.

Biological Psychiatry, 38(3), 174–179.

[https://doi.org/10.1016/0006-3223\(94\)00238-X](https://doi.org/10.1016/0006-3223(94)00238-X)

Mendlewicz, J., Dunbar, G. C., & Hoffman, G. (1985). Changes in sleep EEG architecture

during the treatment of depressed patients with mianserin. *Acta Psychiatrica*

Scandinavica, 72(s320), 26–29. <https://doi.org/10.1111/j.1600-0447.1985.tb08071.x>

Miao, X.-R., Chen, Q.-B., Wei, K., Tao, K.-M., & Lu, Z.-J. (2018). Posttraumatic stress

disorder: From diagnosis to prevention. *Military Medical Research*, 5(1), 32.

<https://doi.org/10.1186/s40779-018-0179-0>

Miller, K. E., & Gehrman, P. R. (2019). REM Sleep: What Is It Good For? *Current Biology*,

29(16), R806–R807. <https://doi.org/10.1016/j.cub.2019.06.074>

Mitchell, H. A., & Weinshenker, D. (2010). Good night and good luck: Norepinephrine in

sleep pharmacology. *Biochemical Pharmacology*, 79(6), 801–809.

<https://doi.org/10.1016/j.bcp.2009.10.004>

Mondino, A., Hambrecht-Wiedbusch, V. S., Li, D., York, A. K., Pal, D., González, J.,

Tortorolo, P., Mashour, G. A., & Vanini, G. (2021a). Glutamatergic Neurons in the

Preoptic Hypothalamus Promote Wakefulness, Destabilize NREM Sleep, Suppress

REM Sleep, and Regulate Cortical Dynamics. *The Journal of Neuroscience: The*

Official Journal of the Society for Neuroscience, 41(15), 3462–3478.

<https://doi.org/10.1523/JNEUROSCI.2718-20.2021>

- Mondino, A., Hambrecht-Wiedbusch, V. S., Li, D., York, A. K., Pal, D., González, J., Torterolo, P., Mashour, G. A., & Vanini, G. (2021b). Glutamatergic Neurons in the Preoptic Hypothalamus Promote Wakefulness, Destabilize NREM Sleep, Suppress REM Sleep, and Regulate Cortical Dynamics. *Journal of Neuroscience*, *41*(15), 3462–3478. <https://doi.org/10.1523/JNEUROSCI.2718-20.2021>
- Morigaki, R., Mure, H., Kaji, R., Nagahiro, S., & Goto, S. (2016). Therapeutic Perspective on Tardive Syndrome with Special Reference to Deep Brain Stimulation. *Frontiers in Psychiatry*, *7*. <https://doi.org/10.3389/fpsyt.2016.00207>
- Murkar, A. L. A., & De Koninck, J. (2018). Consolidative mechanisms of emotional processing in REM sleep and PTSD. *Sleep Medicine Reviews*, *41*, 173–184. <https://doi.org/10.1016/j.smr.2018.03.001>
- Nelson, E. C., Agrawal, A., Pergadia, M. L., Lynskey, M. T., Todorov, A. A., Wang, J. C., Todd, R. D., Martin, N. G., Heath, A. C., Goate, A. M., Montgomery, G. W., & Madden, P. A. F. (2009). Association of childhood trauma exposure and GABRA2 polymorphisms with risk of posttraumatic stress disorder in adults. *Molecular Psychiatry*, *14*(3), 234–235. <https://doi.org/10.1038/mp.2008.81>
- Nollet, M., Wisden, W., & Franks, N. P. (2020). Sleep deprivation and stress: A reciprocal relationship. *Interface Focus*, *10*(3), 20190092. <https://doi.org/10.1098/rsfs.2019.0092>
- Olf, M. (2017). Sex and gender differences in post-traumatic stress disorder: An update. *European Journal of Psychotraumatology*, *8*(sup4), 1351204. <https://doi.org/10.1080/20008198.2017.1351204>
- Olsson, C. A., Anney, R. J. L., Lotfi-Miri, M., Byrnes, G. B., Williamson, R., & Patton, G. C. (2005). Association between the COMT Val158Met polymorphism and propensity to

- anxiety in an Australian population-based longitudinal study of adolescent health. *Psychiatric Genetics*, 15(2), 109.
- Pagel, J. F., & Parnes, B. L. (2001). Medications for the Treatment of Sleep Disorders: An Overview. *Primary Care Companion to the Journal of Clinical Psychiatry*, 3(3), 118–125. <https://doi.org/10.4088/pcc.v03n0303>
- Paiva, H. S., Filho, I. J. Z., & Cais, C. F. D. S. (2021). Using Prazosin to Treat Posttraumatic Stress Disorder and Associations: A Systematic Review. *Psychiatry Investigation*, 18(5), 365–372. <https://doi.org/10.30773/pi.2020.0411>
- Perry, B. D., Giller, E. L., & Southwick, S. M. (1987). Altered platelet alpha 2-adrenergic binding sites in posttraumatic stress disorder. *The American Journal of Psychiatry*, 144(11), 1511–1512. <https://doi.org/10.1176/ajp.144.11.1511a>
- Pitman, R. K. (1989). Post-traumatic stress disorder, hormones, and memory. *Biological Psychiatry*, 26(3), 221–223. [https://doi.org/10.1016/0006-3223\(89\)90033-4](https://doi.org/10.1016/0006-3223(89)90033-4)
- Pitman, R. K., Rasmusson, A. M., Koenen, K. C., Shin, L. M., Orr, S. P., Gilbertson, M. W., Milad, M. R., & Liberzon, I. (2012a). Biological Studies of Posttraumatic Stress Disorder. *Nature Reviews. Neuroscience*, 13(11), 769–787. <https://doi.org/10.1038/nrn3339>
- Pitman, R. K., Rasmusson, A. M., Koenen, K. C., Shin, L. M., Orr, S. P., Gilbertson, M. W., Milad, M. R., & Liberzon, I. (2012b). Biological Studies of Posttraumatic Stress Disorder. *Nature Reviews. Neuroscience*, 13(11), 769–787. <https://doi.org/10.1038/nrn3339>

- Pless Kaiser, A., Cook, J. M., Glick, D. M., & Moye, J. (2019). Posttraumatic Stress Disorder in Older Adults: A Conceptual Review. *Clinical Gerontologist*, 42(4), 359–376.
<https://doi.org/10.1080/07317115.2018.1539801>
- Plotsky, P. M. (1987). Facilitation of immunoreactive corticotropin-releasing factor secretion into the hypophysial-portal circulation after activation of catecholaminergic pathways or central norepinephrine injection. *Endocrinology*, 121(3), 924–930.
<https://doi.org/10.1210/endo-121-3-924>
- Pooley, A. E., Benjamin, R. C., Sreedhar, S., Eagle, A. L., Robison, A. J., Mazei-Robison, M. S., Breedlove, S. M., & Jordan, C. L. (2018a). Sex differences in the traumatic stress response: PTSD symptoms in women recapitulated in female rats. *Biology of Sex Differences*, 9, 31. <https://doi.org/10.1186/s13293-018-0191-9>
- Pooley, A. E., Benjamin, R. C., Sreedhar, S., Eagle, A. L., Robison, A. J., Mazei-Robison, M. S., Breedlove, S. M., & Jordan, C. L. (2018b). Sex differences in the traumatic stress response: PTSD symptoms in women recapitulated in female rats. *Biology of Sex Differences*, 9, 31. <https://doi.org/10.1186/s13293-018-0191-9>
- Pruiksma, K. E., Taylor, D. J., Wachen, J. S., Mintz, J., Young-McCaughan, S., Peterson, A. L., Yarvis, J. S., Borah, E. V., Dondanville, K. A., Litz, B. T., Hembree, E. A., & Resick, P. A. (2016). Residual sleep disturbances following PTSD treatment in active duty military personnel. *Psychological Trauma: Theory, Research, Practice, and Policy*, 8(6), 697–701. <https://doi.org/10.1037/tra0000150>
- Raskind, M. A., Peterson, K., Williams, T., Hoff, D. J., Hart, K., Holmes, H., Homas, D., Hill, J., Daniels, C., Calohan, J., Millard, S. P., Rohde, K., O'Connell, J., Pritzl, D., Feiszli, K., Petrie, E. C., Gross, C., Mayer, C. L., Freed, M. C., ... Peskind, E. R. (2013). A trial of prazosin for combat trauma PTSD with nightmares in active-duty

- soldiers returned from Iraq and Afghanistan. *The American Journal of Psychiatry*, 170(9), 1003–1010. <https://doi.org/10.1176/appi.ajp.2013.12081133>
- Rasmusson, A. M., Hauger, R. L., Morgan, C. A., Bremner, J. D., Charney, D. S., & Southwick, S. M. (2000). Low baseline and yohimbine-stimulated plasma neuropeptide Y (NPY) levels in combat-related PTSD. *Biological Psychiatry*, 47(6), 526–539. [https://doi.org/10.1016/s0006-3223\(99\)00185-7](https://doi.org/10.1016/s0006-3223(99)00185-7)
- Raza, Z., Hussain, S. F., Foster, V. S., Wall, J., Coffey, P. J., Martin, J. F., & Gomes, R. S. M. (2023). Exposure to war and conflict: The individual and inherited epigenetic effects on health, with a focus on post-traumatic stress disorder. *Frontiers in Epidemiology*, 3. <https://www.frontiersin.org/articles/10.3389/fepid.2023.1066158>
- Richards, A., Kanady, J. C., & Neylan, T. C. (2020). Sleep disturbance in PTSD and other anxiety-related disorders: An updated review of clinical features, physiological characteristics, and psychological and neurobiological mechanisms. *Neuropsychopharmacology*, 45(1), 55–73. <https://doi.org/10.1038/s41386-019-0486-5>
- Riga, D., Kooij, K. L., Rademakers, K., Wolterink-Donselaar, I. G., Basak, O., & Meye, F. J. (2025). Neuropeptide Y neurons surrounding the locus coeruleus inhibit noradrenergic system activity to reduce anxiety. *Science advances*, 11(30), eadq0011. <https://doi.org/10.1126/sciadv.adq0011>
- Roffwarg, H. P., Muzio, J. N., & Dement, W. C. (1966). Ontogenetic development of the human sleep-dream cycle. *Science (New York, N.Y.)*, 152(3722), 604–619. <https://doi.org/10.1126/science.152.3722.604>
- Sabban, E. L., Alaluf, L. G., & Serova, L. I. (2016). Potential of neuropeptide Y for preventing or treating post-traumatic stress disorder. *Neuropeptides*, 56, 19–24. <https://doi.org/10.1016/j.npep.2015.11.004>

- Sah, R., & Geraciotti, T. D. (2013). Neuropeptide Y and posttraumatic stress disorder. *Molecular Psychiatry*, 18(6), 646–655. <https://doi.org/10.1038/mp.2012.101>
- Sareen, J. (2014). Posttraumatic stress disorder in adults: Impact, comorbidity, risk factors, and treatment. *Canadian Journal of Psychiatry. Revue Canadienne De Psychiatrie*, 59(9), 460–467. <https://doi.org/10.1177/070674371405900902>
- Schittecatte, M., Dumont, F., Machowski, R., Cornil, C., Lavergne, F., & Wilmotte, J. (2002). Effects of Mirtazapine on Sleep Polygraphic Variables in Major Depression. *Neuropsychobiology*, 46(4), 197–201. <https://doi.org/10.1159/000067812>
- Sealey, J. E., & Laragh, J. H. (2009). Using Plasma Renin (PRA) Testing to Design Follow-Up Drug Treatment Strategies in Hypertensive Patients Already Taking Antirenin System Drugs. *American Journal of Hypertension*, 22(9), 950–950. <https://doi.org/10.1038/ajh.2009.123>
- Shen, Y.-C., Sun, X., Li, L., Zhang, H.-Y., Huang, Z.-L., & Wang, Y.-Q. (2022). Roles of Neuropeptides in Sleep-Wake Regulation. *International Journal of Molecular Sciences*, 23(9), 4599. <https://doi.org/10.3390/ijms23094599>
- Sherin, J. E., & Nemeroff, C. B. (2011a). Post-traumatic stress disorder: The neurobiological impact of psychological trauma. *Dialogues in Clinical Neuroscience*, 13(3), 263–278.
- Sherin, J. E., & Nemeroff, C. B. (2011b). Post-traumatic stress disorder: The neurobiological impact of psychological trauma. *Dialogues in Clinical Neuroscience*, 13(3), 263–278.
- Sherin, J. E., & Nemeroff, C. B. (2011c). Post-traumatic stress disorder: The neurobiological impact of psychological trauma. *Dialogues in Clinical Neuroscience*, 13(3), 263–278. <https://doi.org/10.31887/DCNS.2011.13.2/jsherin>

- Smith, S. M., & Vale, W. W. (2006). The role of the hypothalamic-pituitary-adrenal axis in neuroendocrine responses to stress. *Dialogues in Clinical Neuroscience*, 8(4), 383–395.
- Southwick, S. M., Bremner, J. D., Rasmusson, A., Morgan, C. A., Arnsten, A., & Charney, D. S. (1999). Role of norepinephrine in the pathophysiology and treatment of posttraumatic stress disorder. *Biological Psychiatry*, 46(9), 1192–1204.
[https://doi.org/10.1016/s0006-3223\(99\)00219-x](https://doi.org/10.1016/s0006-3223(99)00219-x)
- Southwick, S. M., Krystal, J. H., Morgan, C. A., Johnson, D., Nagy, L. M., Nicolaou, A., Heninger, G. R., & Charney, D. S. (1993a). Abnormal noradrenergic function in posttraumatic stress disorder. *Archives of General Psychiatry*, 50(4), 266–274.
<https://doi.org/10.1001/archpsyc.1993.01820160036003>
- Southwick, S. M., Krystal, J. H., Morgan, C. A., Johnson, D., Nagy, L. M., Nicolaou, A., Heninger, G. R., & Charney, D. S. (1993b). Abnormal noradrenergic function in posttraumatic stress disorder. *Archives of General Psychiatry*, 50(4), 266–274.
<https://doi.org/10.1001/archpsyc.1993.01820160036003>
- Spindler, M., Palombo, M., Zhang, H., & Thiel, C. M. (2023). Dysfunction of the hypothalamic-pituitary adrenal axis and its influence on aging: The role of the hypothalamus. *Scientific Reports*, 13(1), 6866.
<https://doi.org/10.1038/s41598-023-33922-5>
- Stevens, J. S., & Jovanovic, T. (2019). Role of social cognition in post-traumatic stress disorder: A review and meta-analysis. *Genes, Brain and Behavior*, 18(1), e12518.
<https://doi.org/10.1111/gbb.12518>

- Suchecki, D., Tiba, P. A., & Machado, R. B. (2012). REM Sleep Rebound as an Adaptive Response to Stressful Situations. *Frontiers in Neurology*, 3, 41. <https://doi.org/10.3389/fneur.2012.00041>
- Tasan, R. O., Verma, D., Wood, J., Lach, G., Hörmer, B., De Lima, T. C. M., Herzog, H., & Sperk, G. (2016). The role of Neuropeptide Y in fear conditioning and extinction. *Neuropeptides*, 55, 111–126. <https://doi.org/10.1016/j.npep.2015.09.007>
- Thomas, M. M., Harpaz-Rotem, I., Tsai, J., Southwick, S. M., & Pietrzak, R. H. (2017). Mental and Physical Health Conditions in US Combat Veterans: Results From the National Health and Resilience in Veterans Study. *The Primary Care Companion for CNS Disorders*, 19(3). <https://doi.org/10.4088/PCC.17m02118>
- Torrise, S. A., Leggio, G. M., Drago, F., & Salomone, S. (2019). Therapeutic Challenges of Post-traumatic Stress Disorder: Focus on the Dopaminergic System. *Frontiers in Pharmacology*, 10, 404. <https://doi.org/10.3389/fphar.2019.00404>
- Valente, N. L. M., Vallada, H., Cordeiro, Q., Bressan, R. A., Andreoli, S. B., Mari, J. J., & Mello, M. F. (2011). Catechol-O-methyltransferase (COMT) val158met Polymorphism as a Risk Factor for PTSD After Urban Violence. *Journal of Molecular Neuroscience*, 43(3), 516–523. <https://doi.org/10.1007/s12031-010-9474-2>
- van der Kolk, B. (2000). Posttraumatic stress disorder and the nature of trauma. *Dialogues in Clinical Neuroscience*, 2(1), 7–22. <https://doi.org/10.31887/DCNS.2000.2.1/bvdkolk>
- Van Erum, J., Van Dam, D., & De Deyn, P. P. (2019b). Alzheimer's disease: Neurotransmitters of the sleep-wake cycle. *Neuroscience & Biobehavioral Reviews*, 105, 72–80. <https://doi.org/10.1016/j.neubiorev.2019.07.019>

- Vanderheyden, W. M., George, S. A., Urpa, L., Kehoe, M., Liberzon, I., & Poe, G. R. (2015). Sleep alterations following exposure to stress predict fear-associated memory impairments in a rodent model of PTSD. *Experimental Brain Research*, 233(8), 2335–2346. <https://doi.org/10.1007/s00221-015-4302-0>
- Vanderheyden, W. M., Poe, G. R., & Liberzon, I. (2014). Trauma exposure and sleep: Using a rodent model to understand sleep function in PTSD. *Experimental Brain Research*, 232(5), 1575–1584. <https://doi.org/10.1007/s00221-014-3890-4>
- Vanini, G., Wathen, B. L., Lydic, R., & Baghdoyan, H. A. (2011). Endogenous GABA levels in the pontine reticular formation are greater during wakefulness than during rapid eye movement sleep. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 31(7), 2649–2656. <https://doi.org/10.1523/JNEUROSCI.5674-10.2011>
- Vermetten, E., & Bremner, J. D. (2002). Circuits and systems in stress. II. Applications to neurobiology and treatment in posttraumatic stress disorder. *Depression and Anxiety*, 16(1), 14–38. <https://doi.org/10.1002/da.10017>
- Vgontzas, A. N., Mastorakos, G., Bixler, E. O., Kales, A., Gold, P. W., & Chrousos, G. P. (1999). Sleep deprivation effects on the activity of the hypothalamic-pituitary-adrenal and growth axes: Potential clinical implications. *Clinical Endocrinology*, 51(2), 205–215. <https://doi.org/10.1046/j.1365-2265.1999.00763.x>
- Vitalis, T., & Verney, C. (2022). Roles of the Serotonergic System in Coping with Traumatic Stress. In B. Olivier (Ed.), *Serotonin and the CNS - New Developments in Pharmacology and Therapeutics*. IntechOpen. <https://doi.org/10.5772/intechopen.97221>

- Wagner, U., Hallschmid, M., Rasch, B., & Born, J. (2006). Brief Sleep After Learning Keeps Emotional Memories Alive for Years. *Biological Psychiatry*, 60(7), 788–790.
<https://doi.org/10.1016/j.biopsych.2006.03.061>
- Wang, H. E., Campbell-Sills, L., Kessler, R. C., Sun, X., Heeringa, S. G., Nock, M. K., Ursano, R. J., Jain, S., & Stein, M. B. (2019). Pre-deployment insomnia is associated with post-deployment post-traumatic stress disorder and suicidal ideation in US Army soldiers. *Sleep*, 42(2). <https://doi.org/10.1093/sleep/zsy229>
- Wichniak, A., Wierzbicka, A., Wałęcka, M., & Jernajczyk, W. (2017). Effects of Antidepressants on Sleep. *Current Psychiatry Reports*, 19(9), 63.
<https://doi.org/10.1007/s11920-017-0816-4>
- Yehuda, R., Brand, S., & Yang, R.-K. (2006). Plasma neuropeptide Y concentrations in combat exposed veterans: Relationship to trauma exposure, recovery from PTSD, and coping. *Biological Psychiatry*, 59(7), 660–663.
<https://doi.org/10.1016/j.biopsych.2005.08.027>
- Yehuda, R., Halligan, S. L., & Bierer, L. M. (2001). Relationship of parental trauma exposure and PTSD to PTSD, depressive and anxiety disorders in offspring. *Journal of Psychiatric Research*, 35(5), 261–270.
[https://doi.org/10.1016/s0022-3956\(01\)00032-2](https://doi.org/10.1016/s0022-3956(01)00032-2)
- Zhang, J., Li, J., Liu, C., Gui, H., Yuan, C., & Zhang, Y. (2022). The role of intracerebral dopamine D1 and D2 receptors in sleep-wake cycles and general anesthesia. *Ibrain*, 8(1), 48–54. <https://doi.org/10.1002/ibra.12024>
- Ziegler, C., Wolf, C., Schiele, M. A., Feric Bojic, E., Kucukalic, S., Sabic Džananovic, E., Goci Uka, A., Hoxha, B., Haxhibeqiri, V., Haxhibeqiri, S., Kravic, N., Muminovic Umihanic, M., Cima Franc, A., Jaksic, N., Babic, R., Pavlovic, M., Warrings, B.,

Bravo Mehmedbasic, A., Rudan, D., ... Domschke, K. (2018). Monoamine Oxidase A Gene Methylation and Its Role in Posttraumatic Stress Disorder: First Evidence from the South Eastern Europe (SEE)-PTSD Study. *International Journal of Neuropsychopharmacology*, 21(5), 423–432. <https://doi.org/10.1093/ijnp/pyx111>